KUNAL GOENKA

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AI Engineer specializing in agentic systems, Model Context Protocol (MCP) integrations, and enterprise RAG across regulated environments like Azure Government Cloud. Experienced in taking ambiguous operational bottlenecks from initial problem discovery to production software, delivering rapid time-to-value with measurable business ROI.

SKILLS

AI & Systems Architecture: Model Context Protocol (MCP), LangGraph, Agentic State Machines, Hybrid Retrieval (BM25 + Dense Vectors), LLM-as-a-Judge Evaluation, Cross-Encoder Reranking, Human-in-the-Loop Routing
Engineering & Languages: Python, FastAPI, TypeScript, React, Java Spring Boot, SQL, REST APIs, Docker, PyTest, CI/CD, Git
Cloud, Data & Observability: Azure Government Cloud, AWS S3, ChromaDB, MySQL, OpenTelemetry (OTel)

PROFESSIONAL EXPERIENCE

The Boeing Company — **Arlington, VA**
Data Scientist – Applied AI Systems — *July 2025 – Present*

Enterprise AI & Data (EAI&D)

- **Architected an enterprise Agentic RAG engine** exposing relational databases via Model Context Protocol (MCP) tool servers and LangGraph state machines; implemented dynamic query decomposition and hybrid search (BM25 + dense vectors), cutting hallucination rates by 35% on an automated LLM-as-a-judge harness.
- **Slashed 95% of manual cycle time** (80 → 4 hours per aircraft program) on Nonconformance Report (NCR) resolutions by deploying an end-to-end entity resolution pipeline pairing deterministic heuristics with transformer-based semantic matching.
- **Engineered an asynchronous defense document ingestion pipeline** (FastAPI, Docker) converting unformatted aerospace PDFs and legacy engineering schematics into structured vector metadata for 4,500+ AWS S3 artifacts at <1.2s P95 retrieval latency.
- **Scaled citation-grounded semantic search across 3,000-page airline maintenance manuals**, building confidence-score routing to automatically escalate ambiguous queries for active-learning review by safety engineers.
- **Directed a 7-person team delivering an Azure Government Cloud intern-recruiting platform** (Spring Boot, Node.js, React) for 630+ hiring managers, using Amazon Kiro's spec-driven agentic-AI workflows to turn requirements into validated hiring, compliance, interview, offer, and Workday-handoff features**.**
- **Received CTO's Performance & Innovation Award** for platform architecture and infrastructure engineering supporting the enterprise AI Summit (6.8K unique users, 24K page views).

ELORA (Employee Listening, Organizational Research, and Assessments)

- **Shipped a high-throughput multilingual NLP** pipeline (fine-tuned RoBERTa + spaCy) to classify topics, sentiment, and operational risk across 100K+ enterprise records in 10+ languages, lifting validation F1 from **0.90 to 0.98**.
- **Architected a production semantic search platform** using ChromaDB vector indexing and custom embeddings, securing endpoints with JWT-based RBAC to enable executive queries across historical datasets with citation validation.

EDUCATION

Georgia Tech: College of Computing — **Atlanta, GA**
Master of Science in Computer Science (OMSCS) – AI Concentration — August 2025 – Present

- Relevant Coursework: Knowledge Based AI (KBAI)

Virginia Tech: Pamplin College of Business — **Blacksburg, VA**
Bachelor of Science in Business Information Technology – Decision Support Systems (DSS) — Graduation Date: May 2025